

Fig. 2: An actual-size, single-side PCB for touch-tone dialer

to 8-ohm speaker LS1 to produce the mixed tone.

Table 1 shows the frequencies of tones selected through particular rows and columns. Although these frequencies are not the exact frequencies of the touch telephone network, these are well within the tolerance specified by the telephone system. A 3.579MHz crystal (X_{TAL1}) serves as a clock to the circuit. The circuit is powered by a 9V PP3 battery (BATT.1).

Construction and testing

An actual-size, single-side PCB for the touch-tone dialer is shown in Fig. 2 and its component layout in Fig. 3. Suitable connectors are provided on the PCB for connecting the speaker and the power supply. After assembling the circuit on PCB, enclose it in a suitable plastic case.

To test the circuit for proper functioning, connect the 9V battery as shown in Fig. 1 and press switch S13

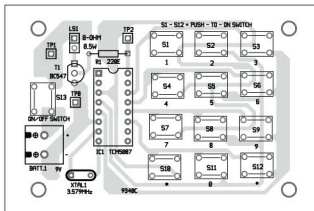


Fig. 3: Component layout for the PCB

to power on the circuit. When you press any key (S1-S12), you will hear the corresponding audio tone through speaker LS1. Now, if you place a telephone handset near speaker LS1 and dial any number, you will see the number getting dialed through the telephone, if the tones are correctly generated. Use test points mentioned in the test points table to debug the circuit, if required. ●

The author is a regular contributor to EPC

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